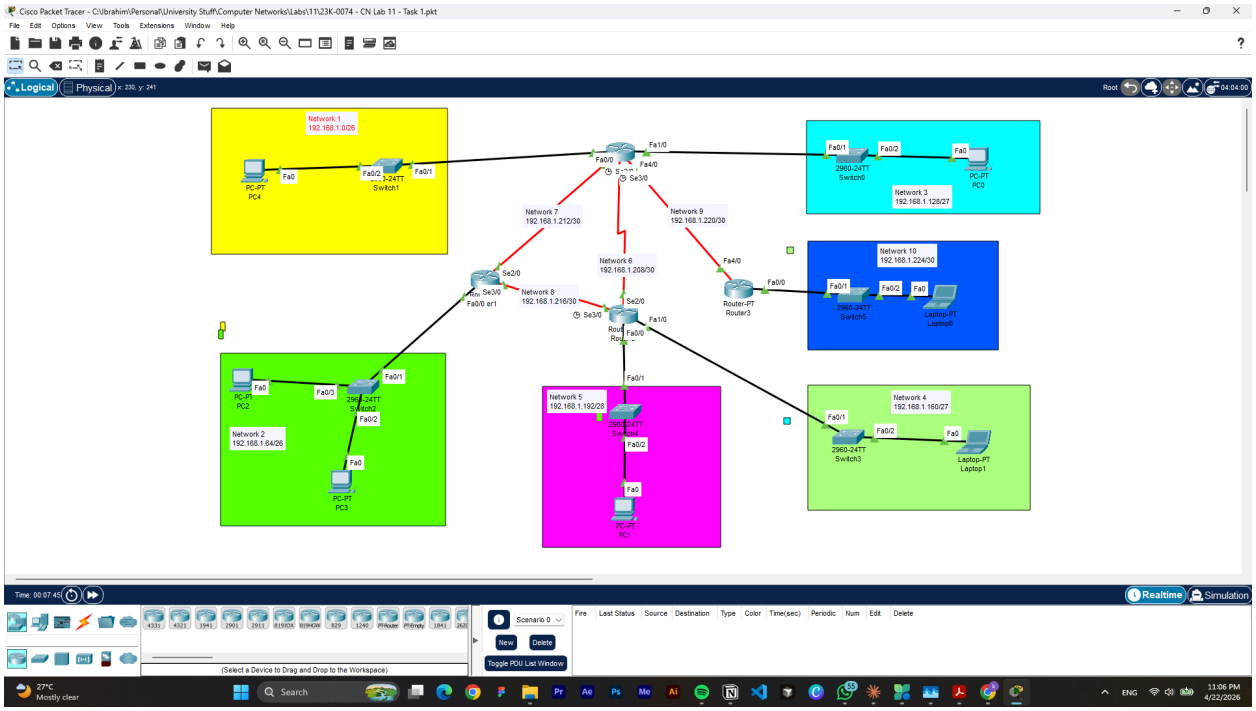
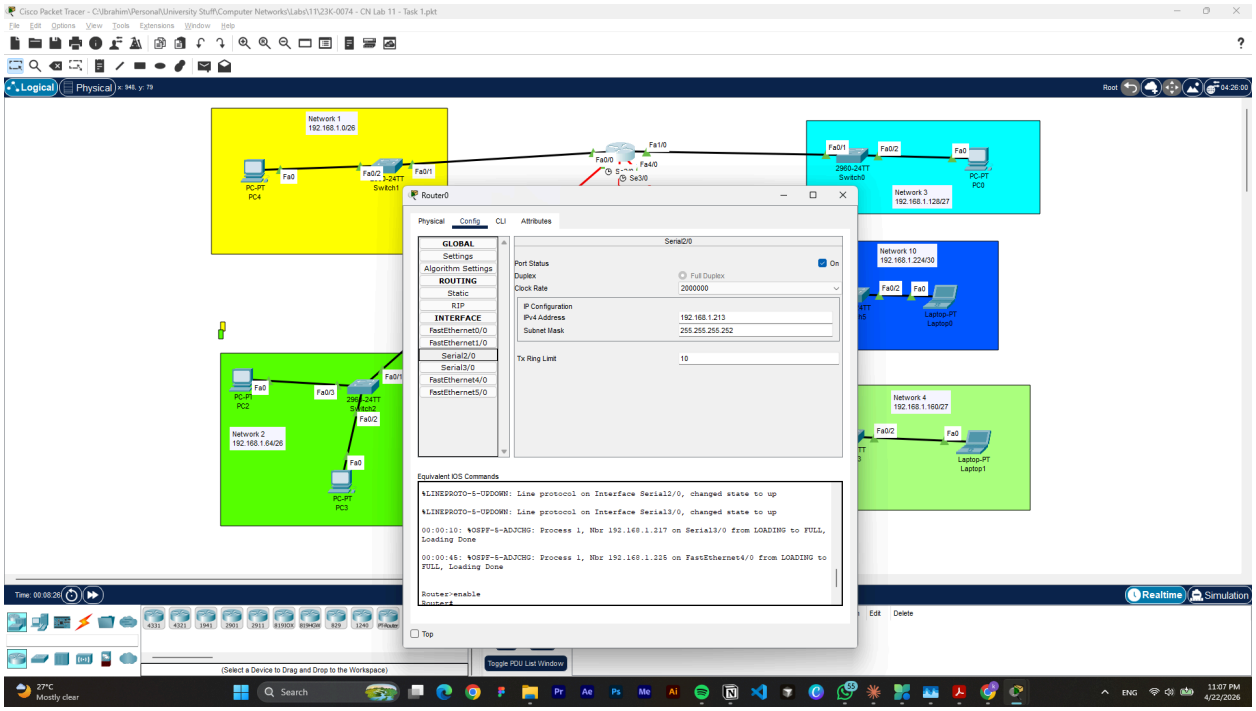
Ibrahim Johar Farooqi
BAI-6A
23K-0074
CN Lab - Lab 11 Tasks

Task 01





Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
         --- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]:

Press RETURN to get started!



Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip address 192.168.1.65 255.255.255.192
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#int fa1/0
Router(config-if)#exit
Router(config)#int se2/0
Router(config-if)#ip address 192.168.1.214
% Incomplete command.
Router(config-if)#ip address 192.168.1.214 255.255.255.252
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

Router(config-if)#exit
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up
int se3/0
Router(config-if)#ip address 192.168.1.218 255.255.255.252
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial3/0, changed state to down
Router(config-if)#exit
Router(config)#
```

Copy

Router2

Physical Config CLI Attributes

IOS Command Line Interface

```
Press RETURN to get started!



Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip address 192.168.1.193 255.255.255.240
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
exit
Router(config)#int fa1/0
Router(config-if)#ip addres 192.168.1.161 255.255.255.224
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up
exit
Router(config)#int se2/0
Router(config-if)#ip address 192.168.1.209 255.255.255.252
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up
exit
Router(config)#int
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up

% Incomplete command.
Router(config)#int se3/0
Router(config-if)#ip address 192.168.1.217 255.255.255.252
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial3/0, changed state to up
```

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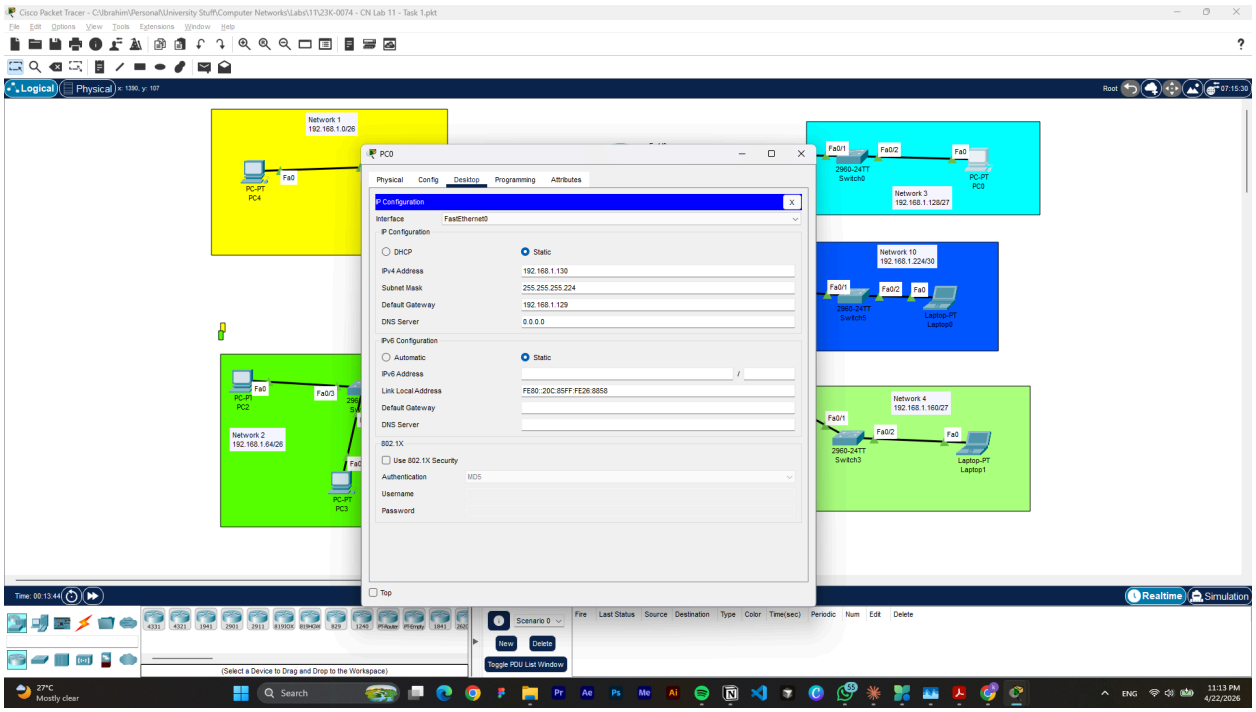
Router3

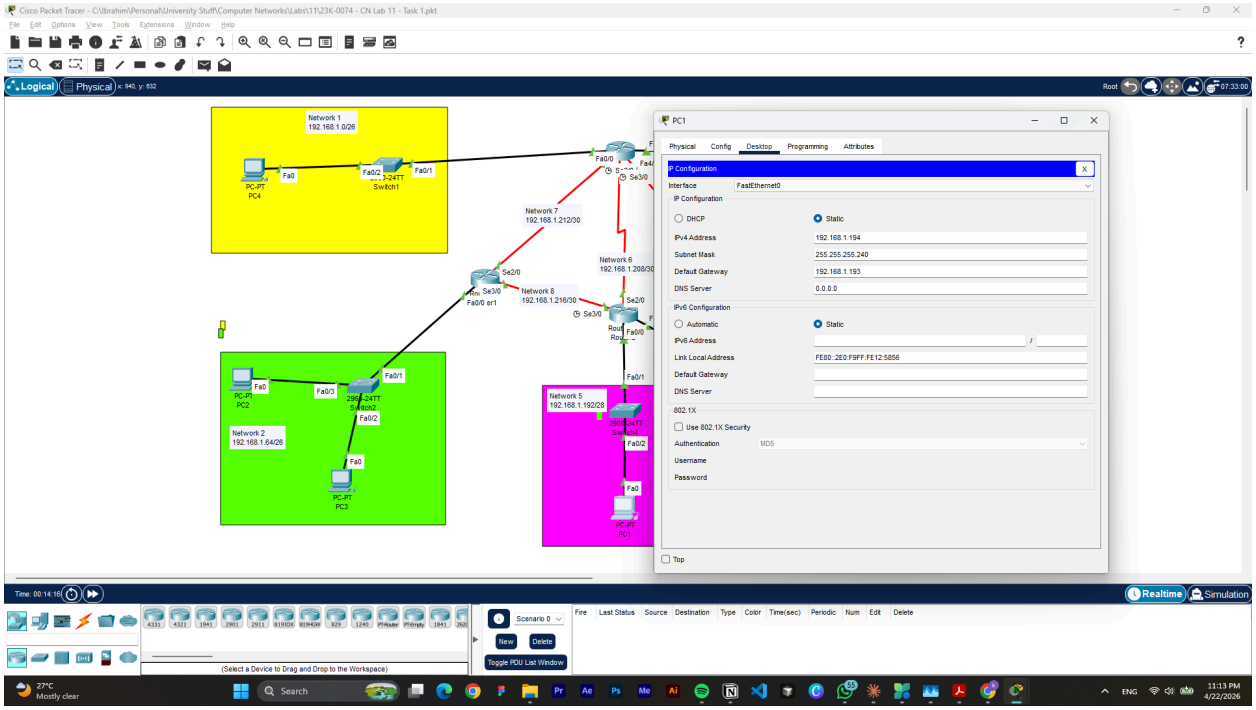
Physical Config CLI Attributes

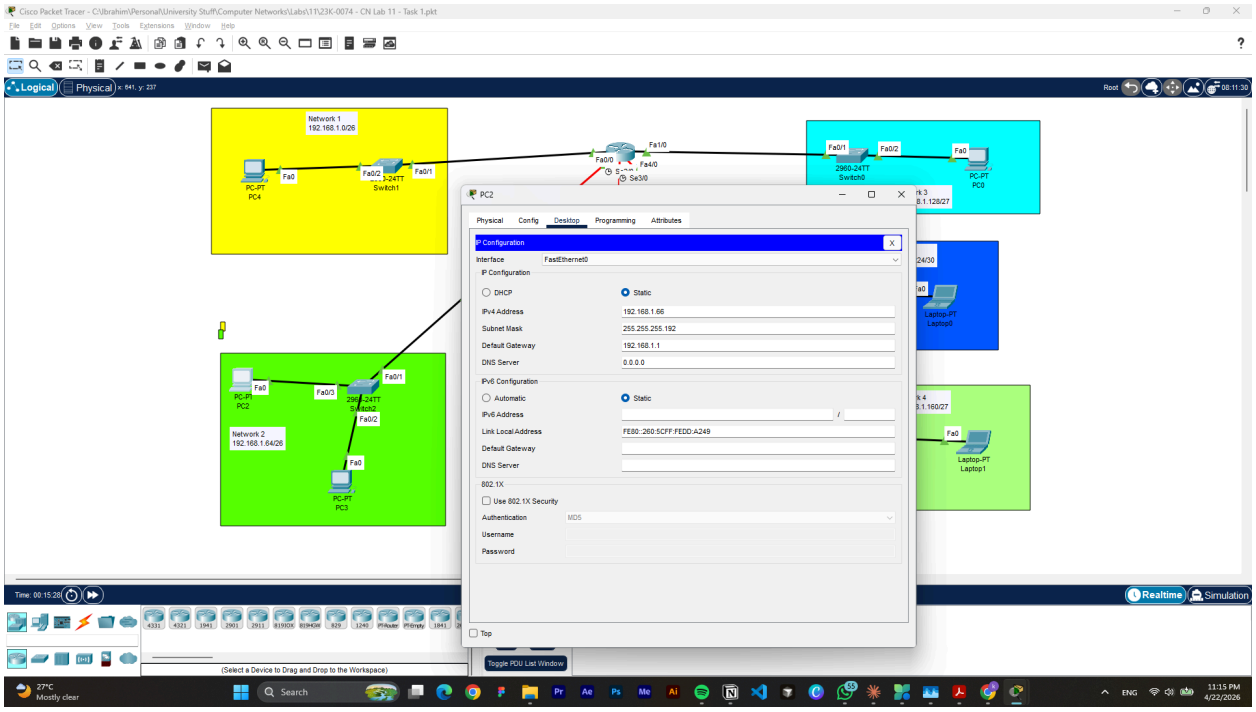
IOS Command Line Interface

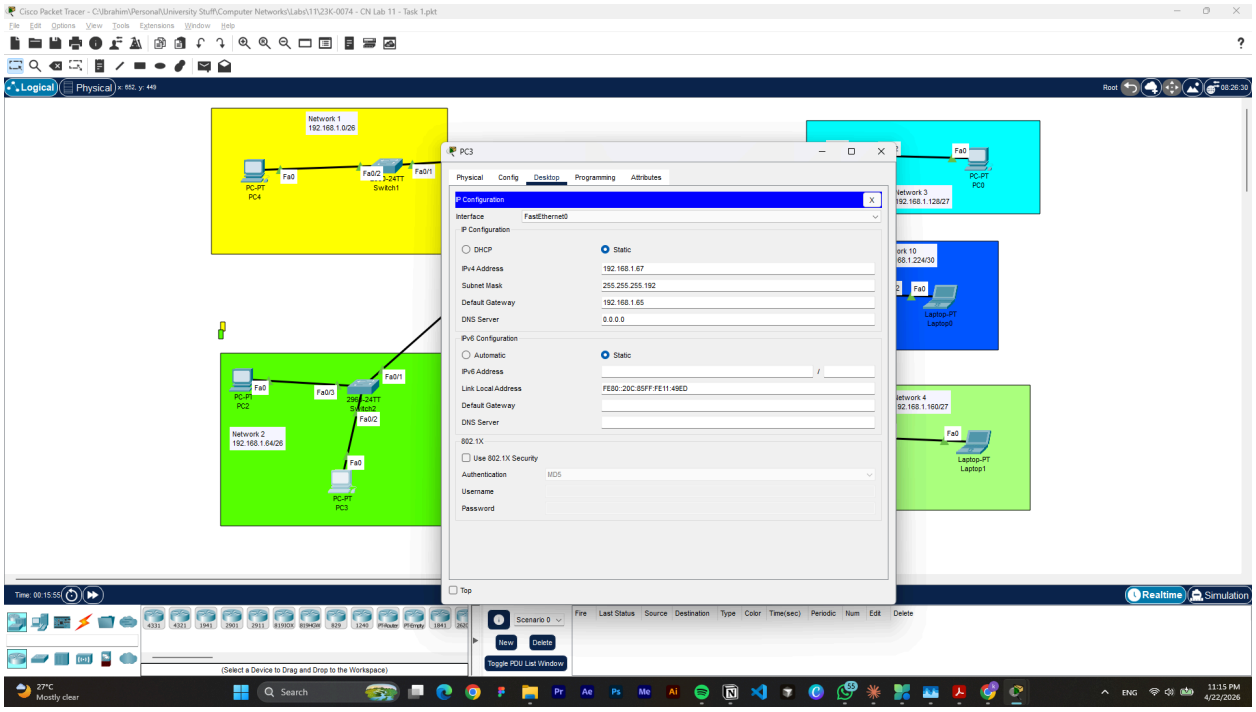
```
Compiled Wed 27-Apr-04 19:01 by miwang

PT 1001 (PTSC2005) processor (revision 0x200) with 60416K/5120K bytes of memory
.
Processor board ID PT0123 (0123)
PT2005 processor: part number 0, mask 01
Bridging software.
X.25 software, Version 3.0.0.
4 FastEthernet/IEEE 802.3 interface(s)
2 Low-speed serial(sync/async) network interface(s)
32K bytes of non-volatile configuration memory.
63488K bytes of ATA CompactFlash (Read/Write)


         --- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]:

Press RETURN to get started!



Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip address 192.168.1.225 255.255.255.252
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#int fa4/0
Router(config-if)#ip address 192.168.1.222 255.255.255.252
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet4/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet4/0, changed state to up
exit
Router(config)#
```

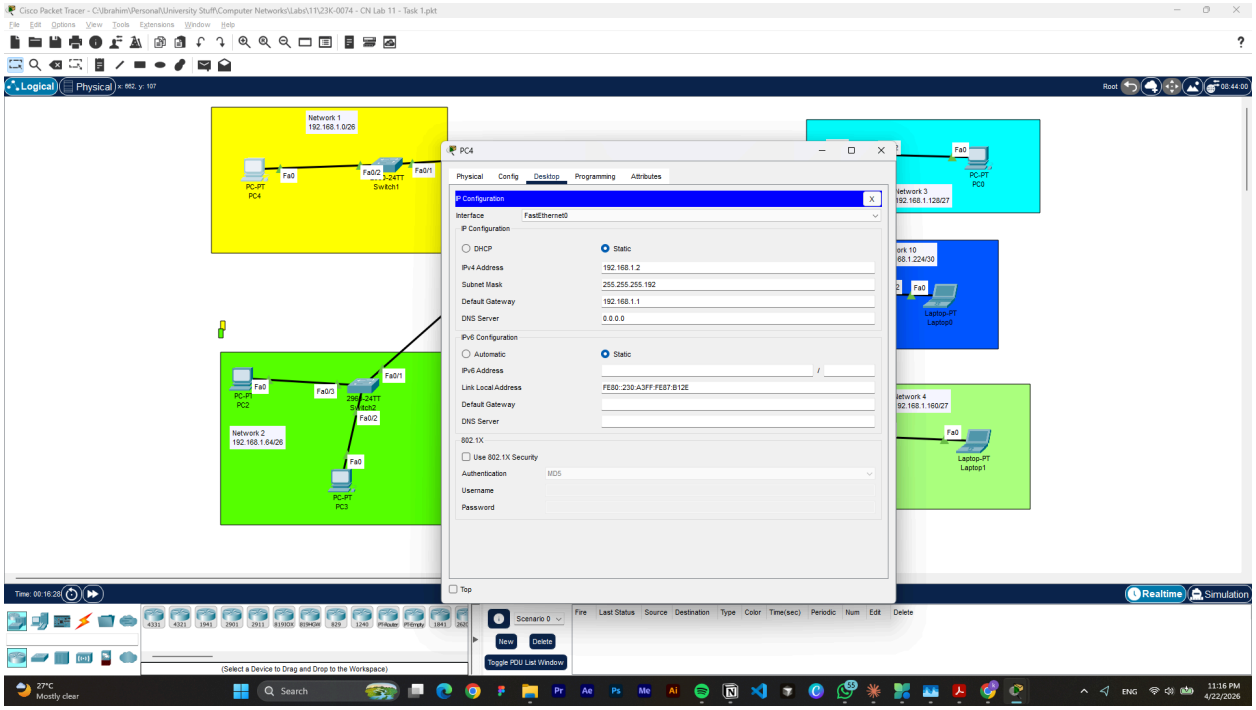
Copy

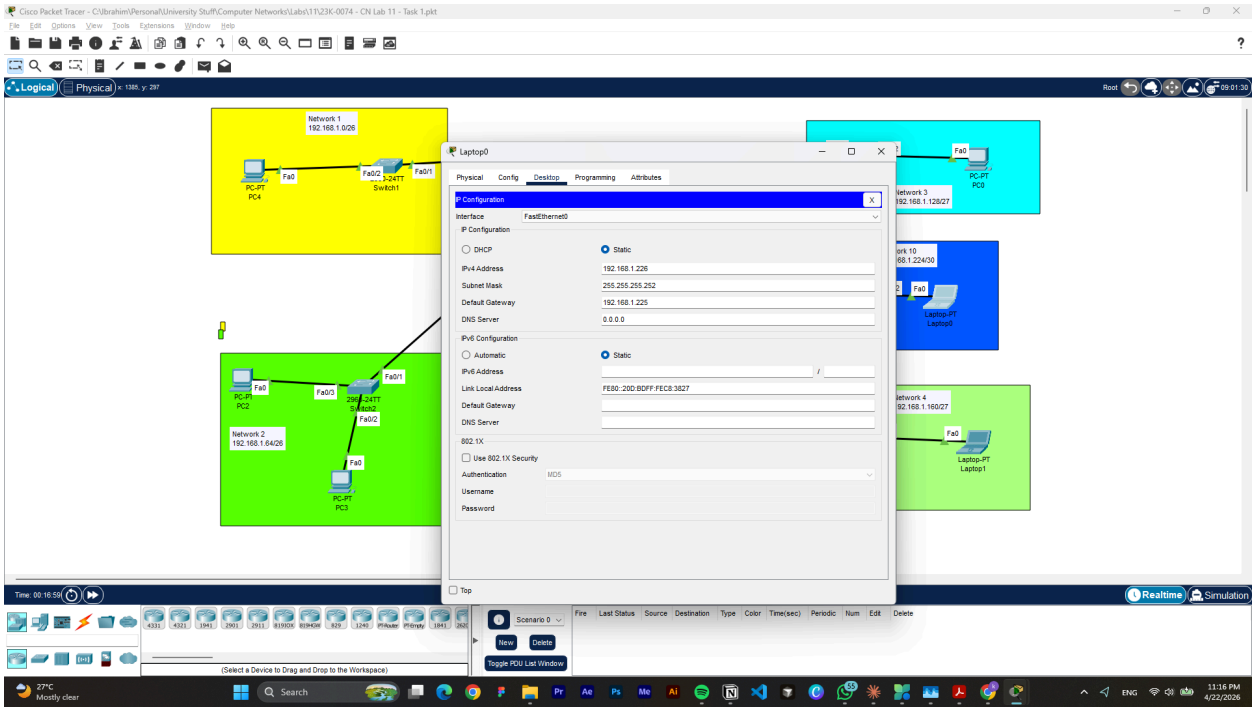


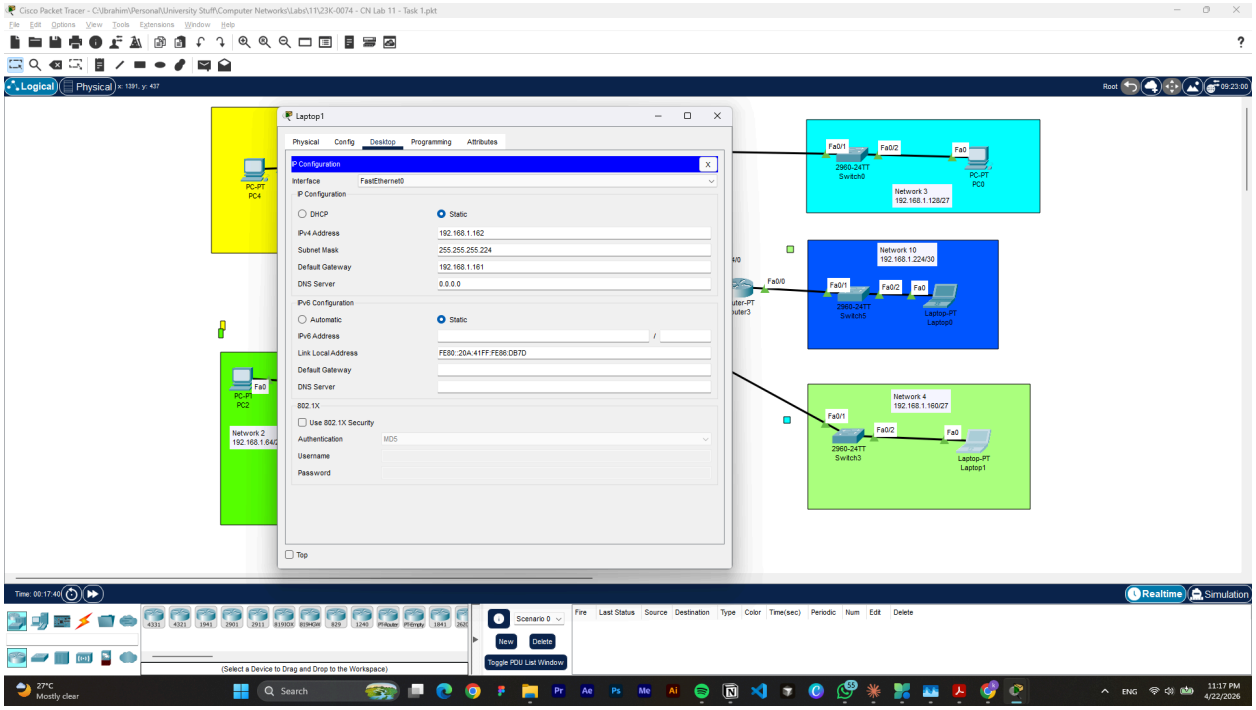

Network 1
192.168.1.0/26
Network 2
192.168.1.64/26
Network 5
192.168.1.192/28
Network 6
192.168.1.208/30
Network 7
192.168.1.212/30
Network 8
192.168.1.216/30
PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
DHCP
Static
IPv4 Address
192.168.1.194
Subnet Mask
255.255.255.240
Default Gateway
192.168.1.193
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::2E0:F9FF:FE12:5856
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top


Network 1
192.168.1.0/26
Network 2
192.168.1.64/26
PC2
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
DHCP
Static
IPv4 Address
192.168.1.66
Subnet Mask
255.255.255.192
Default Gateway
192.168.1.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::260:5CFF:FEDD:A249
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top


Cisco Packet Tracer - C:\Ibrahim\Personal\University Stuff\Computer Networks\Labs\11\23K-0074 - CN Lab 11 - Task 1.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
Network 1
192.168.1.0/26
PC-PT
PC4
Fa0
Fa0/2
2960-24TT
Switch1
Fa0/1
PC-PT
PC0
Network 3
192.168.1.128/27
Network 10
192.168.1.224/30
Laptop-PT
Laptop0
Network 4
192.168.1.160/27
Laptop-PT
Laptop1
PC-PT
PC2
Fa0/3
Switch2
Network 2
192.168.1.64/26
PC-PT
PC3
PC3
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
DHCP
Static
IPv4 Address
192.168.1.67
Subnet Mask
255.255.255.192
Default Gateway
192.168.1.65
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::20C:85FF:FE11:49ED
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete
(Select a Device to Drag and Drop to the Workspace)
27°C
Mostly clear
Search
ENG
11:15 PM
4/22/2026


Cisco Packet Tracer - C:\Ibrahim\Personal\University Stuff\Computer Networks\Labs\11\23K-0074 - CN Lab 11 - Task 1.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
Network 1
192.168.1.0/26
PC-PT
PC4
Fa0
Fa0/2
2960-24TT
Switch1
Fa0/1
PC-PT
PC0
Network 3
192.168.1.128/27
Network 10
192.168.1.224/30
Laptop-PT
Laptop0
Network 4
192.168.1.160/27
Laptop-PT
Laptop1
PC-PT
PC2
Fa0/3
Switch2
Network 2
192.168.1.64/26
PC-PT
PC3
PC4
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
DHCP
Static
IPv4 Address
192.168.1.2
Subnet Mask
255.255.255.192
Default Gateway
192.168.1.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::230:A3FF:FE87:B12E
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete
(Select a Device to Drag and Drop to the Workspace)
27°C
Mostly clear
Search
ENG
11:16 PM
4/22/2026


Cisco Packet Tracer - C:\Ibrahim\Personal\University Stuff\Computer Networks\Labs\11\23K-0074 - CN Lab 11 - Task 1.pkt
Network 1
192.168.1.0/26
PC-PT
PC4
Fa0
Fa0/2
2-24TT
Switch1
Fa0/1
Laptop0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.226
Subnet Mask
255.255.255.252
Default Gateway
192.168.1.225
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::20D:BDFF:FEC8:3827
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top
Network 3
192.168.1.128/27
PC-PT
PC0
Network 10
192.168.1.224/30
Laptop-PT
Laptop0
Network 4
192.168.1.160/27
Laptop-PT
Laptop1
Network 2
192.168.1.64/26
PC-PT
PC2
Fa0/3
2960-24TT
Switch2
Fa0/1
Fa0/2
PC-PT
PC3
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete
(Select a Device to Drag and Drop to the Workspace)
27°C
Mostly clear
Search
ENG
11:16 PM
4/22/2026


Cisco Packet Tracer - C:\Ibrahim\Personal\University Stuff\Computer Networks\Labs\11\23K-0074 - CN Lab 11 - Task 1.pkt
Laptop1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.162
Subnet Mask
255.255.255.224
Default Gateway
192.168.1.161
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::20A:41FF:FE86:DB7D
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top
PC-PT
PC4
Network 3
192.168.1.128/27
PC-PT
PC0
Network 10
192.168.1.224/30
Laptop-PT
Laptop0
Network 4
192.168.1.160/27
Laptop-PT
Laptop1
Network 2
PC-PT
PC2
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
(Select a Device to Drag and Drop to the Workspace)
27°C
Mostly clear
Search
ENG
11:17 PM
4/22/2026

Enabling OSPF

Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Router con0 is now available

Press RETURN to get started.

Press RETURN to get started!

Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 192.168.1.0 0.0.0.63 area 0
Router(config-router)#network 192.168.1.208 0.0.0.3 area 0
Router(config-router)#network 192.168.1.218 0.0.0.3 area 0
Router(config-router)#network 192.168.1.220 0.0.0.3 area 0
Router(config-router)#network 192.168.1.128 0.0.0.31 area 0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#write memory
Building configuration...
[OK]
Router#
```

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Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
Router con0 is now available

Press RETURN to get started.

Press RETURN to get started!

Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 192.168.1.64 0.0.0.63 area 0
Router(config-router)#network 192.168.1.212 0.0.0.3 area 0
Router(config-router)#network 192.168.1.216 0.0.0.3 area 0
Router(config-router)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
```

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Router2

Physical Config CLI Attributes

IOS Command Line Interface

```
Router con0 is now available



Press RETURN to get started.



Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 192.168.1.192 0.0.0.15 area 0
Router(config-router)#network 192.168.1.208 0.0.0.3 area 0
Router(config-router)#
01:09:49: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.221 on Serial2/0 from LOADING to FULL,
Loading Done

Router(config-router)#network 192.168.1.208 0.0.0.3 area 0
Router(config-router)#network 192.168.1.216 0.0.0.3 area 0
Router(config-router)#
01:10:52: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.218 on Serial3/0 from LOADING to FULL,
Loading Done

Router(config-router)#network 192.168.1.216 0.0.0.3 area 0
Router(config-router)#network 192.168.1.160 0.0.0.31 area 0
Router(config-router)#do wr
Building configuration...
[OK]
Router(config-router)#
```

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Router3

Physical Config CLI Attributes

IOS Command Line Interface

```
Router con0 is now available



Press RETURN to get started.



Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 192.168.1.220 0.0.0.3 area 0
Router(config-router)#network 192.168.1.224 0.0.0.3 area 0
Router(config-router)#
01:12:10: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.221 on FastEthernet4/0 from LOADING to FULL
Loading Done

Router(config-router)#network 192.168.1.224 0.0.0.3 area 0
Router(config-router)#do wr
Building configuration...
[OK]
Router(config-router)#
```

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Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.168.1.0/24 is variably subnetted, 10 subnets, 4 masks
C       192.168.1.0/26 is directly connected, FastEthernet0/0
O       192.168.1.64/26 [110/129] via 192.168.1.209, 00:03:23, Serial3/0
C       192.168.1.128/27 is directly connected, FastEthernet1/0
O       192.168.1.160/27 [110/65] via 192.168.1.209, 00:03:13, Serial3/0
O       192.168.1.192/28 [110/65] via 192.168.1.209, 00:04:35, Serial3/0
C       192.168.1.208/30 is directly connected, Serial3/0
C       192.168.1.212/30 is directly connected, Serial2/0
O       192.168.1.216/30 [110/128] via 192.168.1.209, 00:03:33, Serial3/0
C       192.168.1.220/30 is directly connected, FastEthernet4/0
O       192.168.1.224/30 [110/2] via 192.168.1.222, 00:02:04, FastEthernet4/0

Router#show ip protocols

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 192.168.1.221
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    192.168.1.0 0.0.0.63 area 0
    192.168.1.208 0.0.0.3 area 0
    192.168.1.216 0.0.0.3 area 0
    192.168.1.220 0.0.0.3 area 0
    192.168.1.128 0.0.0.31 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
    192.168.1.217        110      00:03:33
    192.168.1.218        110      00:03:48
    192.168.1.221        110      00:02:20
    192.168.1.225        110      00:02:20
```

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default administrative distance (AD) for OSPF = 110

PC1

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.130

Pinging 192.168.1.130 with 32 bytes of data:

Reply from 192.168.1.130: bytes=32 time=1ms TTL=126
Reply from 192.168.1.130: bytes=32 time=19ms TTL=126
Reply from 192.168.1.130: bytes=32 time=1ms TTL=126
Reply from 192.168.1.130: bytes=32 time=5ms TTL=126

Ping statistics for 192.168.1.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 19ms, Average = 6ms

C:\>ping 192.168.1.66

Pinging 192.168.1.66 with 32 bytes of data:

Reply from 192.168.1.66: bytes=32 time=1ms TTL=126
Reply from 192.168.1.66: bytes=32 time=1ms TTL=126
Reply from 192.168.1.66: bytes=32 time=29ms TTL=126
Reply from 192.168.1.66: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.1.66:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 29ms, Average = 8ms

C:\>ping 192.168.1.162

Pinging 192.168.1.162 with 32 bytes of data:

Reply from 192.168.1.162: bytes=32 time<1ms TTL=127
Reply from 192.168.1.162: bytes=32 time=1ms TTL=127
Reply from 192.168.1.162: bytes=32 time<1ms TTL=127
Reply from 192.168.1.162: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.1.162:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Top

Laptop0

Physical Config Desktop Programming Attributes

Command Prompt

```
Ping statistics for 192.168.1.67:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.1.67

Pinging 192.168.1.67 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.67: bytes=32 time=24ms TTL=124
Reply from 192.168.1.67: bytes=32 time=2ms TTL=124
Reply from 192.168.1.67: bytes=32 time=2ms TTL=124

Ping statistics for 192.168.1.67:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 24ms, Average = 9ms

C:\>ping 192.168.1.162

Pinging 192.168.1.162 with 32 bytes of data:

Reply from 192.168.1.162: bytes=32 time=1ms TTL=125
Reply from 192.168.1.162: bytes=32 time=11ms TTL=125
Reply from 192.168.1.162: bytes=32 time=2ms TTL=125
Reply from 192.168.1.162: bytes=32 time=8ms TTL=125

Ping statistics for 192.168.1.162:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 11ms, Average = 5ms

C:\>ping 192.168.1.226

Pinging 192.168.1.226 with 32 bytes of data:

Reply from 192.168.1.226: bytes=32 time=10ms TTL=128
Reply from 192.168.1.226: bytes=32 time=22ms TTL=128
Reply from 192.168.1.226: bytes=32 time=1ms TTL=128
Reply from 192.168.1.226: bytes=32 time=8ms TTL=128

Ping statistics for 192.168.1.226:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 22ms, Average = 10ms

C:\>
```

Top

PC2

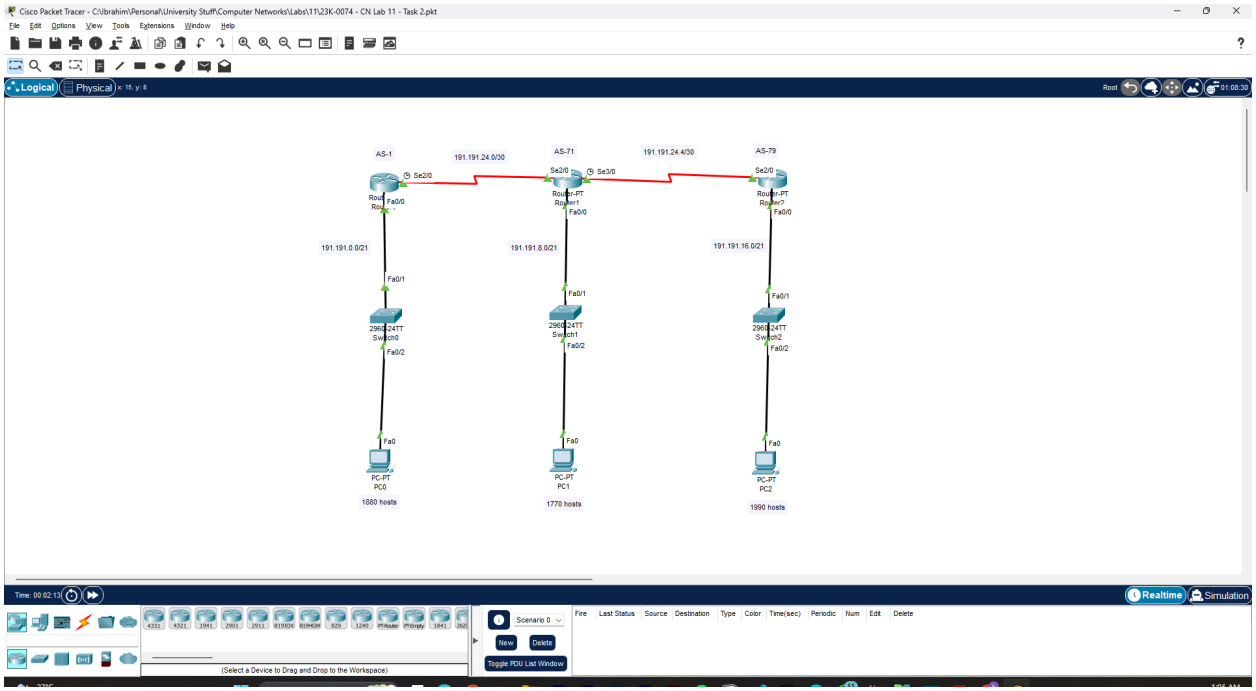
Physical Config Desktop Programming Attributes

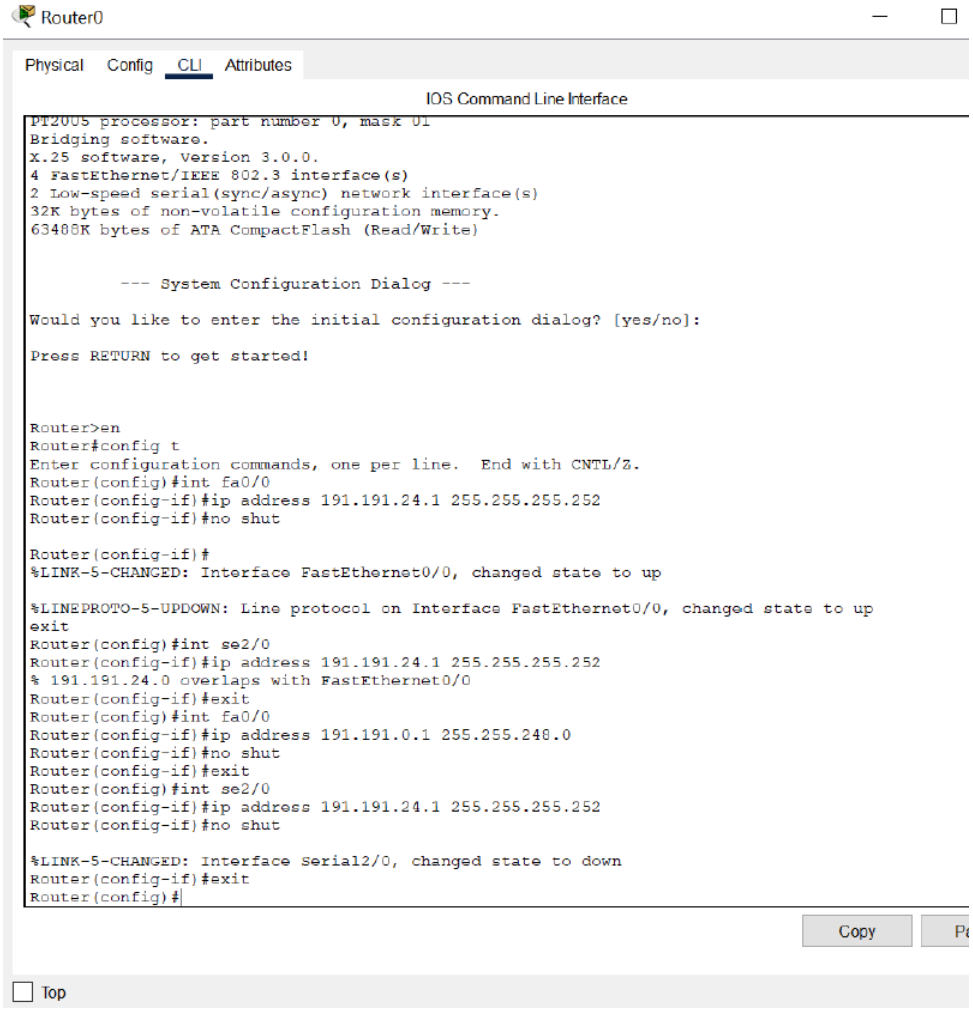
Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=21ms TTL=125
Reply from 192.168.1.2: bytes=32 time=23ms TTL=125
Reply from 192.168.1.2: bytes=32 time=12ms TTL=125
Reply from 192.168.1.2: bytes=32 time=2ms TTL=125

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 23ms, Average = 14ms

C:\>
```

Top

Task 2





Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
PT2005 processor: part number 0, mask 01
Bridging software.
X.25 software, Version 3.0.0.
4 FastEthernet/IEEE 802.3 interface(s)
2 Low-speed serial(sync/async) network interface(s)
32K bytes of non-volatile configuration memory.
63488K bytes of ATA CompactFlash (Read/Write)


         --- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]:

Press RETURN to get started!



Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip address 191.191.24.1 255.255.255.252
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
exit
Router(config)#int se2/0
Router(config-if)#ip address 191.191.24.1 255.255.255.252
% 191.191.24.0 overlaps with FastEthernet0/0
Router(config-if)#exit
Router(config)#int fa0/0
Router(config-if)#ip address 191.191.0.1 255.255.248.0
Router(config-if)#no shut
Router(config-if)#exit
Router(config)#int se2/0
Router(config-if)#ip address 191.191.24.1 255.255.255.252
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial2/0, changed state to down
Router(config-if)#exit
Router(config)#
```

Copy

Top

Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
Bridging software.
X.25 software, Version 3.0.0.
4 FastEthernet/IEEE 802.3 interface(s)
2 Low-speed serial(sync/async) network interface(s)
32K bytes of non-volatile configuration memory.
63488K bytes of ATA CompactFlash (Read/Write)


         --- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]:

Press RETURN to get started!



Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip address 191.191.8.1 255.255.248.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
exit
Router(config)#int se2/0
Router(config-if)#ip address 191.191.24.2 255.255.255.252
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up
exit
Router(config)#int se3/0
Router(config-if)#ip addre
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up
ss 191.191.24.5 255.255.255.252
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial3/0, changed state to down
Router(config-if)#exit
Router(config)#
```

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Router2

Physical Config CLI Attributes

IOS Command Line Interface

```
Compiled Wed 27-Apr-04 19:01 by miwang

PT 1001 (PTSC2005) processor (revision 0x200) with 60416K/5120K bytes of memory
.
Processor board ID PT0123 (0123)
PT2005 processor: part number 0, mask 01
Bridging software.
X.25 software, Version 3.0.0.
4 FastEthernet/IEEE 802.3 interface(s)
2 Low-speed serial(sync/async) network interface(s)
32K bytes of non-volatile configuration memory.
63488K bytes of ATA CompactFlash (Read/Write)


         --- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]:

Press RETURN to get started!



Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip address 191.191.16.1 255.255.248.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
exit
Router(config)#int se2/0
Router(config-if)#ip address 191.191.24.6 255.255.255.252
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up
exit
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up
```

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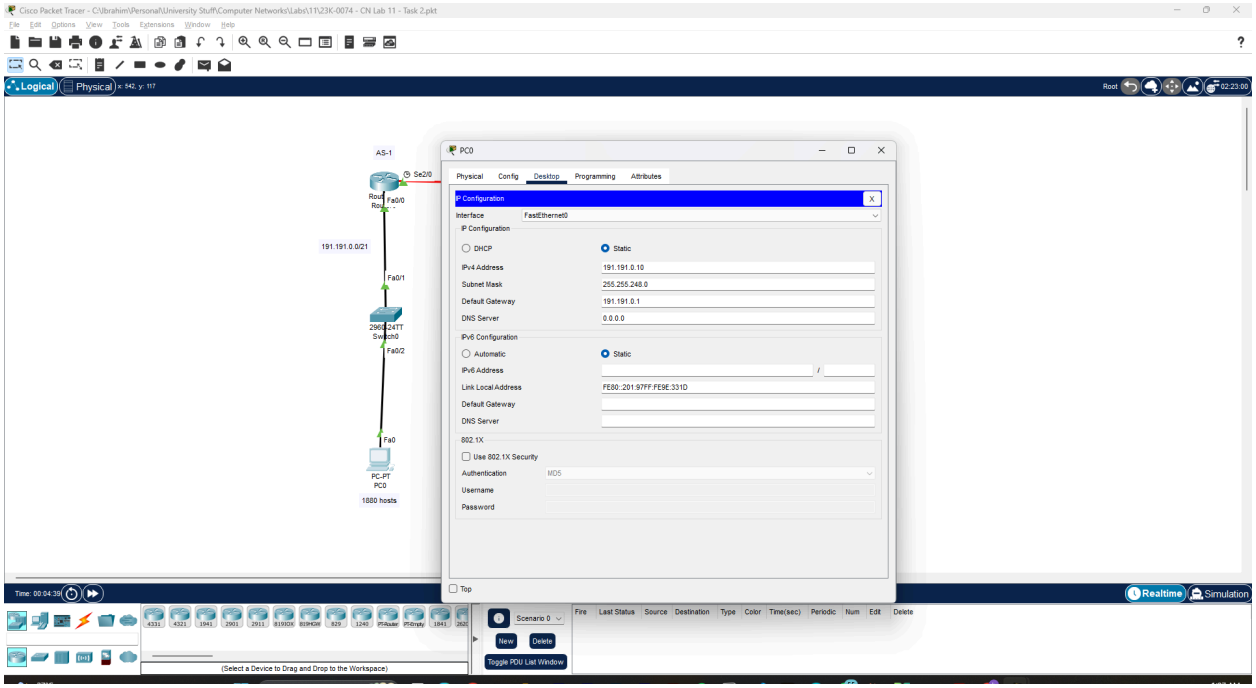
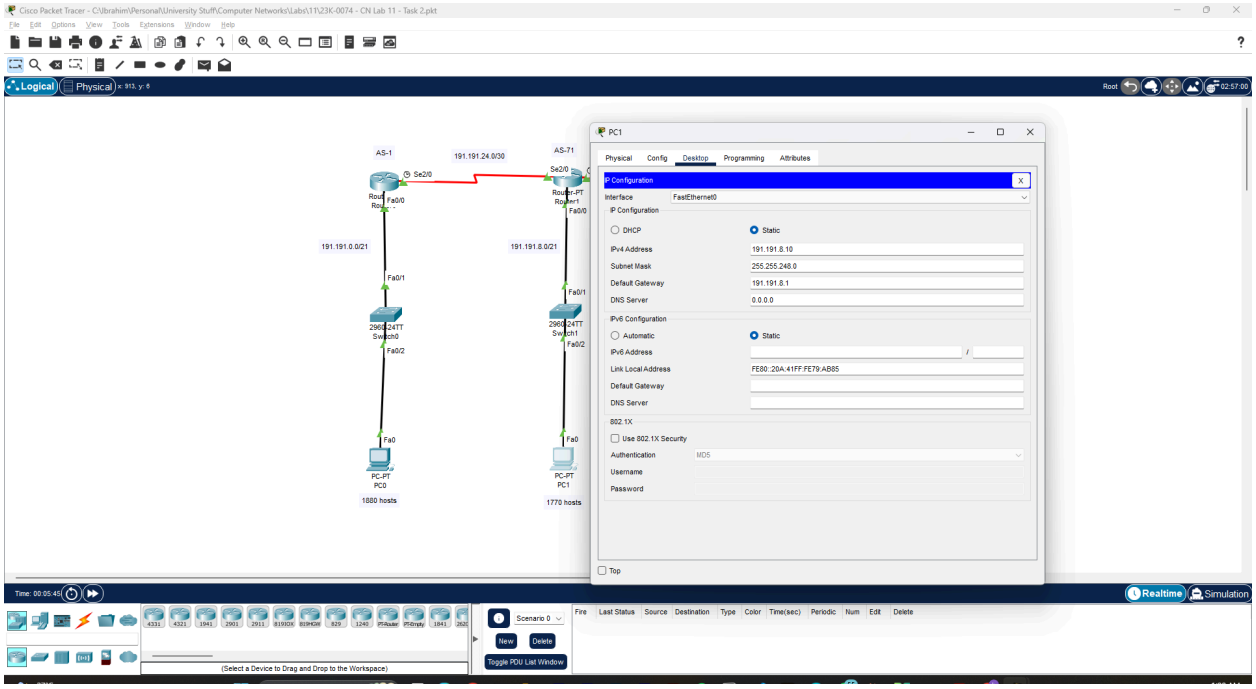
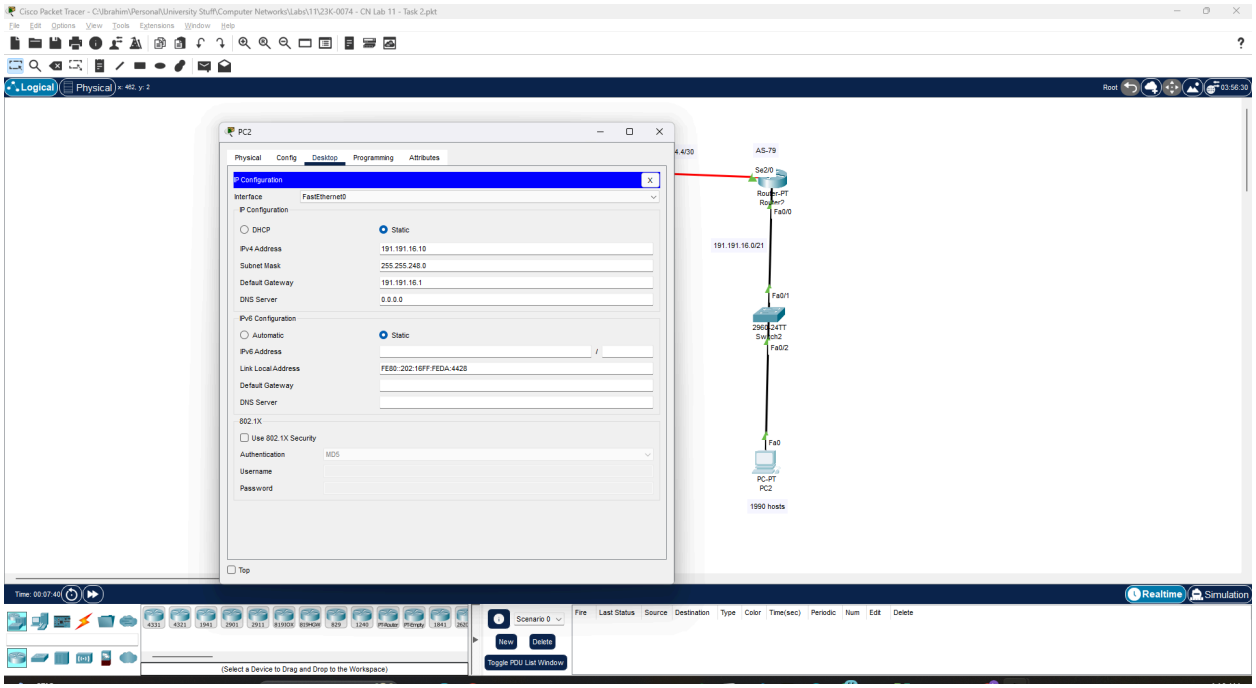
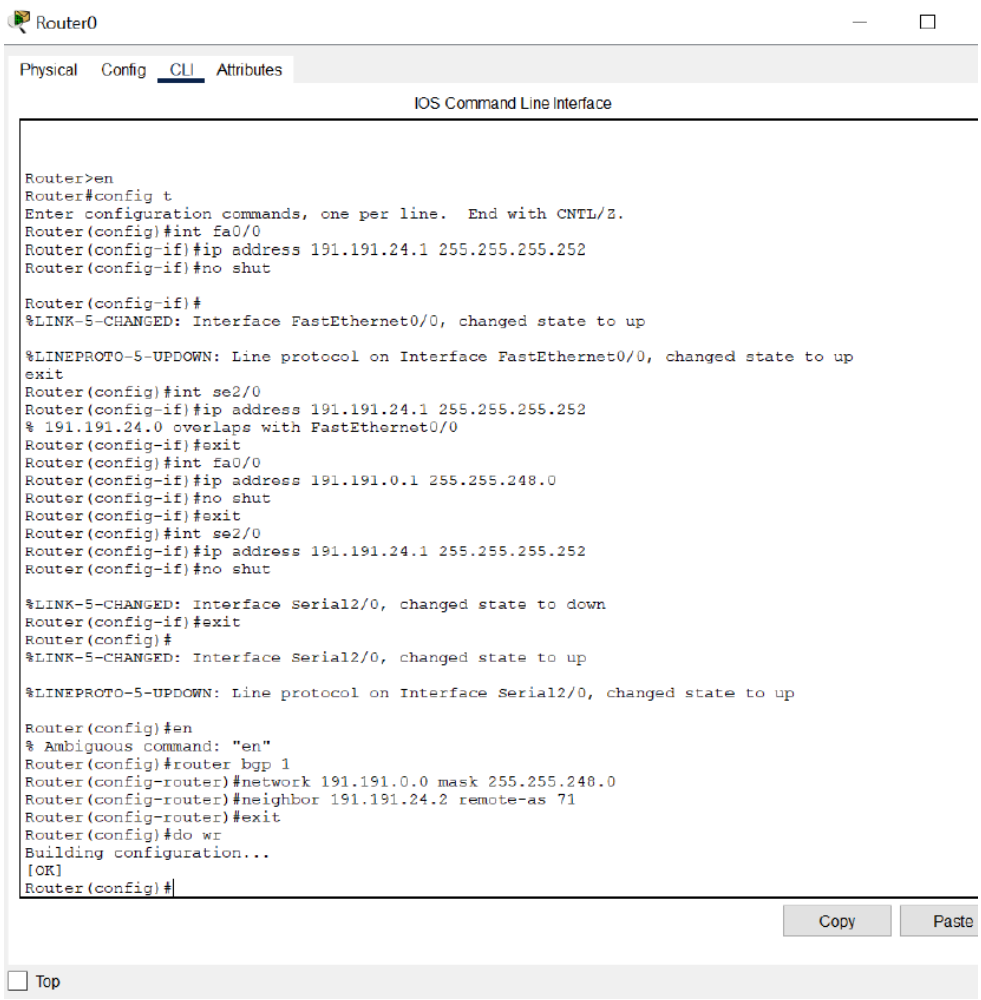
Top


Cisco Packet Tracer - C:\Ibrahim\Personal\University Stuff\Computer Networks\Labs\11\23K-0074 - CN Lab 11 - Task 2.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
AS-1
Se2/0
Fa0/0
191.191.0.0/21
Fa0/1
2960-24TT
Switch0
Fa0/2
Fa0
PC-PT
PC0
1880 hosts
PC0
Physical Config Desktop Programming Attributes
IP Configuration
Interface FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address 191.191.0.10
Subnet Mask 255.255.248.0
Default Gateway 191.191.0.1
DNS Server 0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address FE80::201:97FF:FE9E:331D
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication MD5
Username
Password
Top
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit Delete
(Select a Device to Drag and Drop to the Workspace)


Cisco Packet Tracer - C:\Ibrahim\Personal\University Stuff\Computer Networks\Labs\11\23K-0074 - CN Lab 11 - Task 2.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
AS-1
191.191.24.0/30
AS-71
Se2/0
Fa0/0
191.191.0.0/21
191.191.8.0/21
Fa0/1
2960-24TT
Switch0
Switch1
Fa0/2
Fa0
PC-PT
PC0
1880 hosts
PC-PT
PC1
1770 hosts
PC1
Physical Config Desktop Programming Attributes
IP Configuration
Interface FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address 191.191.8.10
Subnet Mask 255.255.248.0
Default Gateway 191.191.8.1
DNS Server 0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address FE80::20A:41FF:FE79:AB85
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication MD5
Username
Password
Top
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit Delete
(Select a Device to Drag and Drop to the Workspace)



BGP setup



Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip address 191.191.24.1 255.255.255.252
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
exit
Router(config)#int se2/0
Router(config-if)#ip address 191.191.24.1 255.255.255.252
% 191.191.24.0 overlaps with FastEthernet0/0
Router(config-if)#exit
Router(config)#int fa0/0
Router(config-if)#ip address 191.191.0.1 255.255.248.0
Router(config-if)#no shut
Router(config-if)#exit
Router(config)#int se2/0
Router(config-if)#ip address 191.191.24.1 255.255.255.252
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial2/0, changed state to down
Router(config-if)#exit
Router(config)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up

Router(config)#en
% Ambiguous command: "en"
Router(config)#router bgp 1
Router(config-router)#network 191.191.0.0 mask 255.255.248.0
Router(config-router)#neighbor 191.191.24.2 remote-as 71
Router(config-router)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
```

Copy Paste

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Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(config-if)#ip address 191.191.24.2 255.255.255.252
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up
exit
Router(config)#int se3/0
Router(config-if)#ip addre
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed statip address 191.191.24.5
255.255.255.252
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial3/0, changed state to down
Router(config-if)#eiexit
Router(config)#
Router(config)#
Router(config)#interface Serial3/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial2/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#
%LINK-5-CHANGED: Interface Serial3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial3/0, changed state to up

Router(config-if)#exit
Router(config)#router bgp 71
Router(config-router)#network 191.191.8.0 mask 255.255.248.0
Router(config-router)#neighbor 191.191.24.6 remote-as 79
Router(config-router)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#router bgp 71
Router(config-router)#neighbor 191.191.24.1 remote-as 1
Router(config-router)#%BGP-5-ADJCHANGE: neighbor 191.191.24.1 Up

Router(config-router)#do wr
Building configuration...
[OK]
Router(config-router)#
```

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Router2

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up




Router con0 is now available




Press RETURN to get started.


Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router bgp 79
Router(config-router)#network 191.191.16.0 mask 255.255.248.0
Router(config-router)#neighbor 191.191.24.5 remote-as 71
Router(config-router)#%BGP-5-ADJCHANGE: neighbor 191.191.24.5 Up

Router(config-router)#do wr
Building configuration...
[OK]
Router(config-router)#
```

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Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Building configuration...
[OK]
Router(config)#%BGP-5-ADJCHANGE: neighbor 191.191.24.2 Up

Router(config)#show ip route
                ^
% Invalid input detected at '^' marker.

Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     191.191.0.0/16 is variably subnetted, 4 subnets, 2 masks
C       191.191.0.0/21 is directly connected, FastEthernet0/0
B       191.191.8.0/21 [20/0] via 191.191.24.2, 00:00:00
B       191.191.16.0/21 [20/0] via 191.191.24.2, 00:00:00
C       191.191.24.0/30 is directly connected, Serial2/0

Router#show ip protocols
Routing Protocol is "bgp 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  IGP synchronization is disabled
  Automatic route summarization is disabled
  Neighbor(s):
    Address          FiltIn FiltOut DistIn DistOut Weight RouteMap
    191.191.24.2
  Maximum path: 1
    Routing Information Sources:
      Gateway         Distance      Last Update
      191.191.24.2               20         00:00:00
    Distance: external 20 internal 200 local 200
Router#
```

Internal BGP - AD 200 - least preferred

External BGP - AD 20 - highly trusted

PC0

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 191.191.8.10

Pinging 191.191.8.10 with 32 bytes of data:

Request timed out.
Reply from 191.191.8.10: bytes=32 time=1ms TTL=126
Reply from 191.191.8.10: bytes=32 time=1ms TTL=126
Reply from 191.191.8.10: bytes=32 time=6ms TTL=126

Ping statistics for 191.191.8.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 6ms, Average = 2ms

C:\>ping 191.191.8.10

Pinging 191.191.8.10 with 32 bytes of data:

Reply from 191.191.8.10: bytes=32 time=11ms TTL=126
Reply from 191.191.8.10: bytes=32 time=10ms TTL=126
Reply from 191.191.8.10: bytes=32 time=15ms TTL=126
Reply from 191.191.8.10: bytes=32 time=8ms TTL=126

Ping statistics for 191.191.8.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 8ms, Maximum = 15ms, Average = 11ms

C:\>ping 191.191.16.10

Pinging 191.191.16.10 with 32 bytes of data:

Request timed out.
Reply from 191.191.16.10: bytes=32 time=10ms TTL=125
Reply from 191.191.16.10: bytes=32 time=2ms TTL=125
Reply from 191.191.16.10: bytes=32 time=26ms TTL=125

Ping statistics for 191.191.16.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 26ms, Average = 12ms

C:\>
```

Top

PC1

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 191.191.0.10

Pinging 191.191.0.10 with 32 bytes of data:

Reply from 191.191.0.10: bytes=32 time=1ms TTL=126
Reply from 191.191.0.10: bytes=32 time=1ms TTL=126
Reply from 191.191.0.10: bytes=32 time=8ms TTL=126
Reply from 191.191.0.10: bytes=32 time=8ms TTL=126

Ping statistics for 191.191.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 8ms, Average = 4ms

C:\>ping 191.191.16.10

Pinging 191.191.16.10 with 32 bytes of data:

Reply from 191.191.16.10: bytes=32 time=12ms TTL=126
Reply from 191.191.16.10: bytes=32 time=8ms TTL=126
Reply from 191.191.16.10: bytes=32 time=1ms TTL=126
Reply from 191.191.16.10: bytes=32 time=19ms TTL=126

Ping statistics for 191.191.16.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 19ms, Average = 10ms

C:\>
```

PC2

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 191.191.0.10

Pinging 191.191.0.10 with 32 bytes of data:

Reply from 191.191.0.10: bytes=32 time=17ms TTL=125
Reply from 191.191.0.10: bytes=32 time=21ms TTL=125
Reply from 191.191.0.10: bytes=32 time=2ms TTL=125
Reply from 191.191.0.10: bytes=32 time=23ms TTL=125

Ping statistics for 191.191.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 23ms, Average = 15ms

C:\>ping 191.191.8.10

Pinging 191.191.8.10 with 32 bytes of data:

Reply from 191.191.8.10: bytes=32 time=1ms TTL=126
Reply from 191.191.8.10: bytes=32 time=1ms TTL=126
Reply from 191.191.8.10: bytes=32 time=8ms TTL=126
Reply from 191.191.8.10: bytes=32 time=8ms TTL=126

Ping statistics for 191.191.8.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 8ms, Average = 4ms

C:\>
```

Top

191.191.0.0/16

LAN1: 1880 hosts
LAN2: 1770 hosts
LAN3: 1990 hosts

$2^n - 2 >= hosts$

for 1990 hosts = $2^{11} - 2 = 2046$ :: 11 host bits
for 1880 hosts = $2^{11} - 2 = 2046$ :: 11 host bits
for 1770 hosts = $2^{11} - 2 = 2046$ :: 11 host bits

32 - 11 = /21 (for all LANs)

$2^7 + 2^6 + 2^5 + 2^4 + 2^3 = 248$

Subnet mask = 255.255.248.0

Network: 191.191.0.0/21
Range: 191.191.0.1 - 191.191.7.254
Broadcast: 191.191.7.255

Network: 191.191.8.0/21
Range: 191.191.8.1 - 191.191.15.254
Broadcast: 191.191.15.255

Network: 191.191.16.0/21
Range: 191.191.16.1 - 191.191.23.254
Broadcast: 191.191.23.255